



KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPERS**

Website: www.ktunotes.in

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Third Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: MAT201

Course Name: Partial Differential equations and Complex analysis

Duration: 3 Hours

Max. Marks: 100

PART A*Answer all questions. Each question carries 3 marks*

Marks

- 1 Find the partial differential equation by eliminating arbitrary functions f and g from $z = f(x) + g(y)$ (3)
- 2 Solve $\frac{\partial^2 z}{\partial x^2} = xy$ (3)
- 3 Write the three possible solutions of one dimensional wave equation. (3)
- 4 Write any two assumptions used in the derivation of one dimensional heat equation. (3)
- 5 Test the continuity at $z = 0$ of $f(z) = \begin{cases} \frac{Im(z)}{|z|}, & z \neq 0 \\ 0, & z = 0 \end{cases}$ (3)
- 6 Check whether $f(z) = \bar{z}$ is an analytic function? (3)
- 7 Evaluate $\oint_c \frac{e^z}{z-5} dz$, where c is the circle $|z|=4$ (3)
- 8 Find the Taylor series expansion of e^z about $z = \pi$. (3)
- 9 Give example of (a) removable singularity (b) pole (c) essential singularity (3)
- 10 Find the Laurent series expansions of $\frac{1}{z(z-1)}$ about $z = 0$ (3)

PART B*Answer any one full question from each module. Each question carries 14 marks***Module 1**

- 11 (a) Solve $y^2 p - xyq = xz$ 7
- (b) Solve by the method of separation of variables $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$, $u(x, 0) = 4e^{-3x}$ 7
- 12 (a) Find the complete integral of $px + qy = pq$ using Charpit's method. 7
- (b) Form the partial differential equation corresponding to family of spheres with center on z -axis and radius a . 7

Module 2

- 13 (a) Solve the boundary value problem described by $u_{tt}-c^2u_{xx}=0$, $0 \leq x \leq \ell$, $t \geq 0$ 7
 $u(0, t) = u(\ell, t) = 0$, $t \geq 0$, $u(x, 0) = 10 \sin\left(\frac{\pi x}{\ell}\right)$, $\frac{\partial u}{\partial t}(x, 0) = 0$
- (b) Find the temperature $u(x, t)$ in a homogenous bar heat conducting material of length ℓ whose ends kept at 0°C and whose initial temperature is given by 7
 $u(x, 0) = \ell x - x^2$.
- 14 (a) Derive one dimensional wave equation. 7
- (b) The ends A and B of a rod 10 cm in length are kept at temperatures 0°C and 100°C until the steady state condition prevails. If B is Suddenly reduced to 0°C and kept so . Find the temperature distribution in the rod at time t . 7

Module 3

- 15 (a) Show that an analytic function $f(z) = u + iv$ is constant if its modulus is constant. 7
- (b) Find the image of $1 \leq |z| \leq 2$, $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{3}$ under the mapping $w = z^2$ 7
- 16 (a) Verify whether $u = x^3 - 3xy^2$ is harmonic and find its conjugate harmonic function v . 7
- (b) Find the image of the region between real axis and a line parallel to real axis at $y = \frac{\pi}{2}$ under the mapping $W = e^z$ 7

Module 4

- 17 (a) Evaluate $\int_C |z|^2 dz$ where C is the circle $|z| = 2$. 7
- (b) Evaluate $\int_C \frac{z^2+2}{(z-3)^2} dz$ where C is the circle $|z| = 4$ using Cauchy's integral formula 7
- 18 (a) Evaluate $\oint_C \frac{e^z}{(z-1)(z-4)} dz$, where C is $|z| = 2$ using Cauchy's integral formula 7
- (b) Evaluate $\int \frac{3z^2+7z}{z+1} dz$ over (a) $|z| = 1.5$ (b) $|z+i| = 1$ 7

Module 5

- 19 (a) Find the Laurent series expansion for $f(z) = \frac{1}{(z-1)(z-2)}$ valid in 7
 (a) $1 < |z| < 2$ (b) $|z| > 2$

- (b) Evaluate $\int_0^{2\pi} \frac{d\theta}{5-4\sin\theta}$ 7
- 20 (a) Evaluate $\int_{-\infty}^{\infty} \frac{x^2+2}{(x^2+1)(x^2+4)} dx$
- (b) Using residue theorem evaluate $\oint_c \frac{z+1}{z^4-2z^3} dz$, where c is $|z| = \frac{1}{2}$ 7

Ktunotes.in



APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Scheme for Valuation/Answer Key

Scheme of evaluation (marks in brackets) and answers of problems/key
Third Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: MAT201

Course Name: Partial Differential equations and Complex analysis

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks

Marks

- 1 $p = f'(x)$ ------(1) (3)
 $q = g'(y)$ ------(1)
 $s = 0$ ------(1)
- 2 $\frac{\partial z}{\partial x} = \frac{yx^2}{2} + g(y)$ ------(2) (3)
 $z = \frac{yx^3}{6} + x g(y) + f(y)$ ------(1)
- 3 $y = (c_1 e^{\lambda x} + c_2 e^{-\lambda x})(c_3 e^{\lambda at} + c_4 e^{-\lambda at})$ ------(1) (3)
 $y = (c_1 \cos \lambda x + c_2 \sin \lambda x)(c_3 \cos \lambda at + c_4 \sin \lambda at)$ ------(1)
 $y = (c_1 x + c_2)(c_3 t + c_4)$ ------(1)
- 4 Two assumptions ------(2+1) (3)
- 5 For taking path $y = mx$ ------(1) (3)
 Finding $\lim_{z \rightarrow 0} f(z) = \frac{m}{\sqrt{1+m^2}}$ ------(1)
 Conclusion - not continuous------(1)

OR

$$\lim_{z \rightarrow 0} f(z) = \lim_{r \rightarrow 0} \frac{r \sin \theta}{r} = \sin \theta, \text{ Hence not continuous (1+1+1)}$$

- 6 For finding $u_x=1, u_y=0, v_x=0, v_y=-1$ ------(1) (3)
 Cauchy -Riemann equations----- (1)
 CRE not satisfied, hence not analytic------(1)



7 $\frac{e^z}{z-5}$ not analytic at $z=5$ ----- (1) (3)

$z=5$ lies outside $|z|=4$ ----- (1)

By Cauchy's integral theorem $\int \frac{e^z}{z-5} dz = 0$ ----- (1)

8 $e^z = e^{(z-\pi)+\pi}$ ----- (1) (3)

$= e^\pi \left(1 + \frac{z-\pi}{1!} + \frac{(z-\pi)^2}{2!} + \dots \right)$ ----- (2)

OR

Direct method.

Taylor series----- (1)

$f(\pi) = f'(\pi) = f''(\pi) = \dots = e^\pi$ ----- (1)

Substitution & final answer----- (1)

9 Writing any example (1+1+1) (3)

10 $\frac{1}{z(z-1)} = \frac{-1}{z} (1-z)^{-1} = \frac{-1}{z} (1+z+z^2+\dots)$ ----- (1+2) (3)

$f(z) = \frac{1}{z(z-1)} = \frac{1}{z-1} - \frac{1}{z}$ ----- (1)

$= -(1-z)^{-1} - \frac{1}{z} = -\frac{1}{z} - \frac{1}{z} - z - z^2 - \dots$ ----- (1+1)

OR

$f(z) = \frac{1}{z-1} - \frac{1}{z}$ ----- (1)

$= \frac{1}{z(1-\frac{1}{z})} - \frac{1}{z} = \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots \right) - \frac{1}{z}$ ----- (1+1)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1



- 11 (a) Writing AE(Lagrange's subsidiary equations) $\frac{dx}{y^2} = \frac{dy}{-xy} = \frac{dz}{xz}$ ----- 7

(2)

Finding one solution (from the first two ratios) $x^2 + y^2 = c_1$ -----

-(2)

Finding second solution (from the last two ratios) $yz = c_2$ -----

-(2)

Writing general solution $f(x^2 + y^2, yz) = 0$ OR $x^2 + y^2 = f(yz)$

OR $yz = f(x^2 + y^2)$ -----(1)

[*Give suitable stage marks for any other correct attempt.]

- (b) Writing $U = XY$ and finding $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ -----(1) 7

For finding $X = c_1 e^{kx}$ -----(2)

For finding $Y = c_2 e^{-ky}$ -----(2)

Writing $U = C e^{k(x-y)}$ -----(1)

$C = 4, k = -3$ -----(1)

- 12 (a) Writing AE $\frac{dx}{-f_p} = \frac{dy}{-f_q} = \frac{dz}{-pf_p - qf_q} = \frac{dp}{f_x + f_z p} = \frac{dq}{f_y + f_z q}$ ----- 7

(2)

$$f = px + qy - pq;$$

$$f_x = p, f_y = q, f_z = 0, f_p = x - q, f_q = y - p$$

(1)

$$\text{Getting } p = aq$$
 ----- 7

(1)

$$\text{Finding } aq = ax + y \text{ OR } q = x + \frac{y}{a}$$

(1)



$$dz = p dx + q dy \text{ -----}$$

(1)

$$az = \frac{1}{2}(ax + y)^2 + b \text{ -----}$$

(1)

[*Give suitable stage marks for any other correct attempt.]

$$\text{Writing equation of sphere } x^2 + y^2 + (z - c)^2 = a^2 \text{ -----}$$

7

(b) (2)

$$\text{Finding } 2x + 2(z - c)p = 0 \text{ Hence, } z - c = \frac{-x}{p} \text{ -----}$$

(1+1)

$$\text{Finding } 2y + 2(z - c)q = 0 \text{ Hence, } z - c = \frac{-y}{q} \text{ -----}$$

(1+1)

$$\text{Writing PDE } py - qx = 0 \text{ -----}$$

(1)

Module 2



13 (a) $u(x, t) = (A \cos \lambda x + B \sin \lambda x)(C \cos \lambda ct + D \sin \lambda ct)$ -----(1) 7

$$u(0, t) = u(l, t) = 0 \rightarrow A = 0, \lambda = \frac{n\pi}{l}$$
----- (1+1)

$$u(x, t) = \sum_{n=0}^{\infty} \sin\left(\frac{n\pi x}{l}\right) [A_n \cos\left(\frac{n\pi ct}{l}\right) + B_n \sin\left(\frac{n\pi ct}{l}\right)]$$
----- (1)

$$\frac{\partial u}{\partial t}(x, 0) = 0 \rightarrow B_n = 0$$
----- (1)

$$A_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

OR By comparing $10 \sin \frac{\pi x}{l} = \sum_{n=0}^{\infty} A_n \sin\left(\frac{n\pi x}{l}\right)$ ----- (1)

$$A_n = 0, n \neq 1, A_1 = 10$$
 ----- (1)

$$\text{Finding } u(x, t) = 10 \cos \frac{\pi ct}{l} \sin \frac{\pi x}{l}$$

(b)

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$u(x, t) = (A \cos \lambda x + B \sin \lambda x) e^{-\lambda^2 c^2 t}$$
----- (2)

$$u(0, t) = u(l, t) = 0 \rightarrow A = 0, \lambda = \frac{n\pi}{l}$$
----- (1+1)

$$u(x, t) = \sum_{n=0}^{\infty} B_n \sin\left(\frac{n\pi x}{l}\right) \exp\left(-\frac{n^2 \pi^2 c^2 t}{l^2}\right)$$
 ----- (1)

$$B_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$
----- (1)

$$B_n = \frac{4l^2}{n^3 \pi^3} (1 - \cos n\pi) \text{ (OR) } = \frac{4l^2}{n^3 \pi^3} (1 - (-1)^n)$$
----- (1)

14 (a) $F = T(\sin(\varphi + \Delta\varphi) - \sin\varphi)$ ----- (2) 7

$$F = T \left(\frac{\partial y}{\partial x}(x + \Delta x, t) - \frac{\partial y}{\partial x}(x, t) \right)$$
----- (2)

$$\text{By newton's law } F = m \Delta x \frac{\partial^2 y}{\partial t^2}$$
----- (2)

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$
----- (1)

[*Give (2) marks, for writing assumptions only]

(b) Getting Steady state temperature $u = ax + b = 10x$ ----- (1+1) 7

$$u(x, t) = (A \cos \lambda x + B \sin \lambda x) e^{-\lambda^2 c^2 t}$$
----- (1)

$$\text{Boundary conditions } u(0, t) = u(10, t) = 0$$
----- (1)



Applying boundary conditions, $A = 0, \lambda = \frac{n\pi}{10}$ -----(1)

$$u(x, t) = \sum_{n=0}^{\infty} B_n \sin\left(\frac{n\pi x}{10}\right) \exp\left(-\frac{n^2 \pi^2 c^2 t}{100}\right) \text{-----}(1)$$

$$\text{Finding } B_n = -\frac{200}{n\pi} \cos n\pi \text{ (OR)} = \frac{200}{n\pi} (-1)^{n+1} \text{-----}(1)$$

Module 3

- 15 (a) $u^2 + v^2 = c^2$ ----- (1) 7
 $uu_x + vv_x = 0, uu_y + vv_y = 0$ ----- (1+1)
 Writing Cauchy-Riemann equations----- (1)
 $(u_x)^2 + (u_y)^2 = 0$ OR $(u^2 + v^2)u_x = 0$ OR any other similar form---(1)
 u is a constant and/or v is a constant or $f'(z) = 0$ ----- (1)
 Hence $f(z)$ is a constant----- (1)

- (b) Writing $\rho = r^2$ and $\Phi = 2\theta$ ----- (1+1)
 Finding images of $1 \leq |z| \leq 2$ as $1 \leq |w| \leq 4$ ----- (2) 7
 Finding images of $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{3}$ as $\frac{\pi}{3} \leq \Phi \leq \frac{2\pi}{3}$ ----- (2)
 Writing image of given region as $1 \leq |w| \leq 4, \frac{\pi}{3} \leq \Phi \leq \frac{2\pi}{3}$ ----- (1)
 [*Give suitable stage mark for alternate method for finding the images]

- 16 (a) Finding first and second order partial derivatives 7
 $u_x = 3x^2 - 3y^2, u_y = -6xy, u_{xx} = 6x, u_{yy} = -6x$ ----- (1+1)
 Showing u is harmonic----- (2)
 Writing C-R equations ($u_x = v_y$ & $u_y = -v_x$)----- (1)
 Finding $v = 3x^2y - y^3 + c$ ----- (1+1)

OR

Finding first and second order partial derivatives

$$u_x = 3x^2 - 3y^2, u_y = -6xy, u_{xx} = 6x, u_{yy} = -6x \text{-----}(1+1)$$



Showing u is harmonic----- (2)

By Milne-Thomson method, $u_x(z, 0) = 3z^2, u_y(z, 0) = 0$ ----- (1)

$f(z) = \int_C [u_x(z, 0) - iu_y(z, 0)] dz = z^3 + c$ ----- (1)

$v = 3x^2y - y^3 + c$ ----- (1)

(b) Writing $\rho = e^x$ and $\Phi = y$ ----- (1+1)

Finding image of real axis $y = 0$ as $\Phi = 0$ ----- (2)

Finding images of $y = \frac{\pi}{2}$ as $\Phi = \frac{\pi}{2}$ ----- (2)

Writing image of given region as, $0 \leq \Phi \leq \frac{\pi}{2}$, first quadrant ----- (1)

OR

$u = e^x \cos y, v = e^x \sin y$ -----

(1+1)

Finding image of real axis, $y = 0$ as $u = e^x, v = 0$ ----- (2)

Finding images of $y = \frac{\pi}{2}$ as $u = 0, v = e^x$ -----

(2)

Required region is the first quadrant ($u > 0, v > 0$) -----

(1)

Module 4

17 (a) Writing $z = 2e^{i\theta}, 0 \leq \theta \leq 2\pi$ ----- (2)

Finding $dz = 2ie^{i\theta} d\theta, |z|^2 = 4$ ----- (1+1)

$$\int_C |z|^2 dz = \int_0^{2\pi} 4 \times 2ie^{i\theta} d\theta = 8i \frac{e^{i\theta}}{i} \Big|_0^{2\pi} = 0$$

(1+1+1)

(b) $z = 3$ is the singular point----- (2)

Statement of Cauchy's integral formula for derivatives ----- (2)

$f(z) = z^2 + 2$ ----- (1)

$$\int \frac{z^2 + 2}{(z-3)^2} dz = 2\pi i f'(3) = 12\pi i$$



[*Give full marks for correct solution without explicitly writing the statement of CIF]

- 18 (a) $z = 1$ and $z = 4$ are singularities------(2) 7
 only $z = 1$ lies inside given region------(1)
 Statement of Cauchy's integral formula -----(1)
 Here $f(z) = \frac{e^z}{(z-4)}$ ------(1)
 $\int \frac{e^z}{(z-1)(z-4)} = 2\pi i f(1) = 2\pi i \left(\frac{e}{-3}\right)$ ------(1+1)

[*Give full marks for correct solution without explicitly writing the statement of CIF]

- (b) $z = -1$ is a singular point------(1)
 (a) $z = -1$ lies inside $|z| = 1.5$ -----(1)
 By Cauchy's integral formula $\int \frac{3z^2+7z}{z+1} dz = 2\pi i f(-1)$ -----(1) where $f(z)$
 $= 3z^2 + 7z$
 Answer : $-8\pi i$ ------(1)
 (b) $z = -1$ lies outside $|z+i|=1$ OR No singularities inside
 $|z+i|=1$
 -----(1)

Hence by using Cauchy's theorem, $\int \frac{3z^2+7z}{z+1} dz = 0$ ------(1+1)

Module 5



- 19 (a) Converting to partial fractions and writing $f(z) = \frac{-1}{z-1} + \frac{1}{z-2}$ -----(1) (7)

$$1 < |z| < 2 \Rightarrow \frac{1}{|z|} < 1 \text{ and } \frac{|z|}{2} < 1 \text{ -----(1)}$$

$$f(z) = \frac{-1}{z(1-\frac{1}{z})} - \frac{1}{2(1-\frac{z}{2})} \text{ -----(1)}$$

$$= \frac{-1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots \right) - \frac{1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{4} + \dots \right) \text{ -----(1)}$$

$$|z| > 2 \Rightarrow \frac{2}{|z|} < 1 \text{ and } \frac{1}{|z|} < 1 \text{ -----(1)}$$

$$f(z) = \frac{-1}{z(1-\frac{1}{z})} + \frac{1}{z(1-\frac{2}{z})} \text{ -----(1)}$$

$$= -\frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots \right) + \frac{1}{z} \left(1 + \frac{2}{z} + \frac{4}{z^2} + \dots \right) \text{ -----(1)}$$

$$|z|=1, z=e^{i\theta}, dz=ie^{i\theta}d\theta, \sin z = \frac{1}{2i} \left(z - \frac{1}{z} \right) \text{ -----(2)}$$

- (b) Converting integrand as function of z , $\int_C \frac{dz}{5iz-2z^2+2}$ -----(1)

$$\text{Finding singularities } 2i, \frac{1}{2} \text{ -----(1)}$$

$$\text{Finding residue at } \frac{1}{2} = \frac{1}{3i} \text{ -----(1)}$$

$$\text{Hence by Residue theorem } \int_0^{2\pi} \frac{d\theta}{5-4\sin\theta} = \frac{2\pi}{3} \text{ -----(1+1)}$$

- 20 (a) C: the real axis from $-R$ to R and the upper semicircle $|z|=R$ ----- (7)

(1)

$$\text{Converting to complex integrand as } \int_C \frac{z^2+2}{(z^2+1)(z^2+4)} dz \text{ -----}$$

(1)

$$\text{Finding singularities } \pm i, \pm 2i \text{ -----(1)}$$

$$z = i, 2i \text{ lie inside } C \text{ -----(1)}$$

$$\text{Residue at } z = i = \frac{1}{6i} \text{ -----(1)}$$



$$\text{Res}(z = 2i) = \frac{1}{6i} \text{-----}(1)$$

$$\text{Ans } \int_C \frac{x^2+2}{(x^2+1)(x^2+4)} dz = \frac{2\pi}{3} \text{-----}(1)$$

(b) Singularities are $z = 0$ and $z = 2$ -----

(1+1)

Only $z = 0$ is inside given region. It's a pole of order 3-----(1)

$$\text{Res}(z = 0) = \frac{1}{2!} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} z^3 \times \frac{z+1}{z^4-2z^3} = \frac{-3}{8} \text{-----}$$

(1+1)

$$\text{Hence by Residue theorem, } \int_C \frac{z+1}{z^4-2z^3} dz = \frac{-3\pi i}{4} \text{-----}$$

(1+1)

